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Lactation persistency of dairy cows fed low metabolizable protein diets with 2 different starch contents and degradability levels crossed with rumen-protected amino acid supplies

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ABSTRACT

Reducing the CP and MP content of diets allows for improving dairy cow nitrogen and MP efficiencies. However, when CP (≤13%–14%) and MP (<95 g/kg DM of protein digestible in the intestine) contents are reduced, DMI, milk yield, and milk component secretion generally decrease. Improving the balance of nutrient supplies (such as starch or AA) could limit these decreases in milk secretions. These effects have been tested during short-term experiments. The aim of our study was to analyze the effects of increasing the bypass starch content or better balancing AA supplies through rumen-protected AA (RP-AA: Lys, Met, and His) on lactation persistency and milk component secretions when the treatments were applied to Holstein dairy cows from 56 to 183 DIM. Forty-four dairy cows were assigned randomly to 4 groups according to a factorial arrangement of the 4 treatments (LSHDAA−: low-starch, high-degradable in the rumen without any AA supplementation; LSHDAA+: LSHD with RP-AA supplementation; HSLDAA−: high-starch, low-degradable in the rumen without RP-AA supplementation; and HSLDAA+: HSLD with RP-AA supplementation). Increasing the bypass starch content led to increases in milk yield through lactose and milk protein yield with starch × time interactions: lactation persistency improved, as did all milk component secretions in HSLD versus LSHD treatments. Increasing bypass starch slightly increased MP intake but decreased NEL intake. No changes in BCS and plasma insulin were observed when the bypass starch content was increased, probably because NEL content decreased. However, BW increased over time with a tendency to be higher with the higher bypass starch content. In addition, increasing bypass starch or RP-AA intakes increased ME efficiency (i.e., milk energy/ME). Increasing RP-AA supplies increased plasma Lys and Met concentrations, but the concentrations of plasma His only increased on LSHD. Milk protein yield as well as milk protein content and fat yield increased with RP-AA supplementation without any interaction with time. However, significant starch × AA × time interactions were observed for lactose and fat yields. These interactions mainly reflected lower lactation persistency and slopes of milk components in LSHDAA− than under the 3 other treatments. Interestingly, LSHDAA− corresponded to the lowest plasma His concentration compared with LSHDAA+, HSLDAA−, and HSLDAA+. In multiparous cows (n = 32; 8 per dietary treatment), higher mammary cell proliferation, as measured by proliferation cell nuclear antigen staining, was observed with both the HSLD versus LSHD and the AA+ (LSHDAA+ and HSLDAA+) versus AA− (LSHDAA− and HSLDAA−) diets; however, only the 2 HSLD versus LSHD diets gradually increased DMI and consequently whole nutrient supplies to sustain the higher milk persistency. Under a low-MP diet, the LSHDAA− diet proved to be a highly restrictive option for preserving lactation persistency; conversely, increasing the bypass starch content without increasing the NEL content appeared to be a promising solution to counter this negative effect on lactation persistency. Taking account of the type of nutrients (starch or AA and specifically His) absorbed in dairy feeding systems seems important when reducing the MP supply.

Key words: lysine, methionine, histidine, bypass starch, mammary cell turnover

INTRODUCTION

Increasing MP efficiency is important to improve the sustainability and profitability of dairy production. Reducing protein intake enhances nitrogen (N) efficiency

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and reduces protein losses (Kalscheur et al., 2006). However, lowering the dietary MP content below 95 to 100 g/kg DM protein digestible in the intestine (**PDI**; INRA, 2018) decreases both milk protein and milk yields (Daniel et al., 2016), thus affecting profitability for farmers. In practice, low-MP diets are often fed over an extended period, which could reduce lactation persistency after the peak. By lower lactation persistency, we mean a more negative slope of milk yield decline following the peak of lactation. Supplementation with specific nutrients may help to recover these losses in milk and protein yields while still feeding low levels of dietary protein. The availability of some EAA and glucose is crucial to maintain milk production (Gross, 2022). Indeed, many dairy rations do not supply adequate amounts of Met, Lys, and, in some cases, His to fully support the expression of the cows' genetic potential for milk protein synthesis (Schwab and Broderick, 2017; INRA, 2018). It is therefore important to study the effects of specific nutrients on lactation persistency when cows are fed low-protein diets, in terms of N efficiency and the economic profitability of dairy farms (Dekkers et al., 1998).

Balancing the supplies of Lys, Met, and His, the 3 most limiting AA in dairy diets, helps to recover milk true protein yield (**MPY**; Schwab and Broderick, 2017; INRA, 2018; NASEM, 2021), even with a low MP supply (Haque et al., 2012; Lee et al., 2012; CP ≤13% DM). However, their effect on milk yield is less clear. The 2 feeding systems (INRA, 2018; NASEM, 2021) predicted MPY in response to AA supplies (Lys and Met vs. several other EAA, respectively) but did not predict the milk yield response. Increasing His supply has been shown to increase milk yield in mid-lactation cows (Vanhatalo et al., 1999). The increase in milk yield when supplementing with His (Räisänen et al., 2023), Lys or Lys + Met, or Lys + Met + His in low-MP diets may be related to an increase in DMI (Lee et al., 2012; Giallongo et al., 2016). Supplementing diets with Lys and Met could lead to a smaller increase in milk protein content or yield, or both, as lactation progressed (Polan et al., 1991; Socha et al., 2008). Positive effects of these AA on milk yield were observed in cows at the beginning of lactation (Noftsger and St-Pierre, 2003; Robinson et al., 2011), but not consistently (Ordway et al., 2009; Lee et al., 2019; Fehlberg et al., 2020), with no effect noted in mid-lactation (meta-analysis of INRA, 2018). To our knowledge, the effects of supplementing diets with Lys, Met, and His for several months have never before been explored.

Increasing the starch content of the diet can also improve milk and protein yields (Oba and Allen, 2003; Akins et al., 2014; Boerman et al., 2015). Interestingly, more starch also increased MPY under a low MP supply (79 g/kg DM of PDI, equivalent to MP in the INRA (2007) feeding system; CP = 12% DM; Cantalapiedra-Hijar et al., 2014). When a highly fermentable starch source was used, it increased rumen propionate production, which could decrease DMI (Oba and Allen, 2003; Albornoz and Allen, 2018). However, increasing the bypass starch content (starch not degraded in the rumen but potentially digestible in the small intestine and fermented in the hindgut) by using corn grain was able to prevent this DMI decrease (Akins et al., 2014; Boerman et al., 2015). High bypass starch diets may also cause over-fattening in late lactation (NASEM, 2021) when compared with low-bypass starch (Cannas et al., 2002; Oba and Allen, 2003) or lipogenic diets (van Knegsel et al., 2007; Boerman et al., 2015). A higher bypass starch diet may partition more energy toward body gain (lower milk fat yield [**MFY**], increased BW gain, BCS, plasma insulin) in mid- and late-lactation cows (Piccioli-Cappelli et al., 2014).

Combining an increase in bypass starch with Lys, His, and Met supplementation might therefore maintain milk yield under a low-MP supply more effectively than supplementing with these AA alone. To our knowledge, the interaction between starch and rumen-protected Lys, Met, and His supplies has only been analyzed in one short-term study (Zang et al., 2021) under a Latin square design, and they did not observe any interaction. In addition, most studies that were designed to better balance AA or increase the dietary starch content were conducted over short periods of time or presented only overall mean values without analyzing the effects on lactation persistency.

The present study was intended to assess lactation persistency in 44 Holstein dairy cows fed 2 low-MP diets from 56 to 183 DIM that differed in terms of their starch content and its rumen degradability, with or without AA supplementation. We first hypothesized that increasing the content of dietary bypass starch (by using corn starch) without increasing NEL supply could shift energy partitioning toward body reserves rather than milk synthesis at the end of the trial and reduce lactation persistency. We also hypothesized that diets supplemented with AA would increase more milk protein and milk yields at the beginning of the trial than at the end, implying a lower lactation persistency compared with diets not supplemented with AA.

MATERIALS AND METHODS

Animal, Treatments, and Experimental Design

This experiment was carried out at IEPL INRAE Dairy Nutrition and Physiology (Installation Expérimentale de Production Laitière, Institut National de La Recherche Agriculture, Alimentation et Environnement, Le Rheu, France; INRAE, 2021), in accordance with national regulations on animal care (certified by the French Ministry of

Agriculture, Paris; Agreement Number APAFIS#26701-202007231539699-v3), which includes ethical standards in compliance with French law (decree no. 2013-118 February 1, 2013, NOR: AGRG1231951D relating to the protection of animals used for scientific purposes). Forty-four lactating Holstein cows (32 multiparous and 12 primiparous) that calved between September 22 and October 31, 2020, were used in the study. Between 15 and 35 DIM (reference period), the cows produced 34.5 ± 7.68 kg/d milk with protein and fat contents of 30.1 ± 2.03 g/kg and 42.5 ± 5.60 g/kg, respectively, and consumed 21.0 ± 3.67 kg/d DM. The cows were milked twice a day, at 0700 and 1700 h. They were housed in a freestall barn and individually fed. Each cow was assigned an individual feeding trough that was automatically unlocked by an electronic collar.

The cows were assigned to 11 blocks of 4 cows (3 blocks of primiparous and 8 blocks of multiparous cows) according to calving date and mean DMI, milk yield, milk protein and fat yields, milk protein and fat contents, and BCS from 15 to 28 DIM. Within each block, the cows were assigned to 1 of the 4 dietary treatments according to a 2 × 2 factorial design consisting of 2 levels of bypass starch content (low-starch, high-degradable in the rumen [**LSHD**] vs. high-starch, low-degradable in the rumen [**HSLD**]) crossed with 2 levels of Lys, Met, and His supply (without Lys, Met, and His supply [**AA−**] vs. with Lys, Met, and His supply [**AA+**]). The cows then received these treatments continuously from 56 ± 4 to 183 ± 4 DIM (mean ± SD). The experiment started November 17, 2020 (56 d after the first calving date), and finished May 2, 2021 (183 d after the last calving date).

Diets

The animals were individually fed ad libitum (with a 10% refusal rate) a TMR diet twice a day after milking at 0800 and 1600 h, from calving to the end of the experiment. The amount of feed offered was adjusted twice a week, if necessary, on Tuesdays and Fridays to maintain the 10% refusal rate (on a DM basis). The adjustments were calculated on average DMI over the preceding period (Friday–Tuesday or Tuesday–Friday) using the last 7-d mean DM content of each feed ingredient to ensure accurate feed allocation.

From calving to 42 ± 4 DIM, the cows received a diet containing (on a DM basis) 62.4% corn silage, 10.6% alfalfa pellet, 12.7% soybean meal, 12.7% energy concentrate (20% corn, 20% wheat, 20% barley, 20% dehydrated beet pulp, 15% wheat bran, 3% molasses, 1% rapeseed oil, and 1% NaCl), and 1.6% mineral and vitamin premix, as given by Fischer et al. (2018). This diet contained 139 ± 5 g/kg DM of CP, 206 ± 9 g/kg DM of starch, 42 ± 9 g/kg DM of bypass starch, and supplied 1.52 ± 0.03 Mcal/kg DM of NEL and 93 ± 2 g/kg DM of PDI for MP. Based on INRA (2018), feed tables **LysDI**, **MetDI**, and **HisDI** (Lys, Met, or His digestible in the intestine) were estimated to be, respectively, 6.8%, 1.9%, and 2.1% of PDI. From 1 to 15 ± 4 DIM, the cows also received 0.93 ± 0.25 kg/d of DM of chopped (5 cm length) wheat straw mixed into the TMR diet.

From 43 ± 4 to 56 ± 4 DIM (14 d), the cows were gradually switched to the experimental diets (LSHD vs. HSLD) without rumen-protected (**RP**)-AA supplementation. Every 2 d, each feed ingredient from the reference diet was decreased by a constant percentage, while the feed ingredients from the experimental diets were increased also by a constant percentage, so that the final composition of the experimental diets was reached after 14 d. From 56 ± 4 to 183 ± 4 DIM, the 4 groups of cows each received one of the full experimental treatments without (AA−) or with (AA+) RP-AA (**LSHDAA−**, **LSHDAA+**, **HSLDAA−**, and **HSLDAA+**). The feed ingredients in the 4 diets are presented in Table 1, the chemical compositions of the feedstuffs are presented in Table 2, and the chemical compositions of the 4 diets and their final nutritive values are described in Table 3. The LSHD versus HSLD diets were formulated to be close to some diets practiced in northern (LSHD) versus southern (HSLD) Europe, respectively. The LSHD diet was lower in starch but higher in sugar and fermentable OM (**FOM**), which is estimated to enhance VFA production, while the HSLD diet was higher in corn bypass starch. The HSLD diet was formulated to supply +100 g/kg DM of starch (mainly bypass starch through ground corn and corn silage) compared with the LSHD diet. The HSLD diet contained a higher proportion of corn silage than the LSHD diet, and the HSLD concentrate contained corn grain (Tables 1, 2, and 3) to increase bypass starch content compared with LSHD diet. The LSHD diet was composed of chopped grass haylage and corn silage. Molasses was added to the LSHD diets, and the LSHD concentrate contained barley, wheat, and beet pulp (Table 1) to favor VFA production in the rumen but not specifically propionate. The LSHD and HSLD diets were formulated to be isoenergetic on an NEL basis (1.57 Mcal/kg DM of NEL) and to supply a similar NDF content (355 g/kg DM) based on the INRA (2018) feed tables and on preliminary analyses of the feeds. To ensure a similar NDF content, 2 hays were added to the HSLD diets (Tables 1 and 3). The LSHD and HSLD diets were also formulated to be iso-nitrogenous (140 g/kg DM of CP) and iso-MP on a PDI basis (92 g/kg DM of PDI) and have similar contents of LysDI, MetDI, and HisDI (as a percent of PDI; INRA, 2018) under the LSHDAA− and HSLDAA− treatments. Additionally, the rumen protein balance, which indicates RDP and FOM equilibrium (INRA, 2018), should be close to zero. For this reason, alfalfa hay was added to the permanent grassland hay in HSLD diet. The concentrates in the LSHD and HSLD diets contained different proportions of soybean, rapeseed, and corn gluten meal, as well as other ingredients to meet these objectives (Tables 1, 2, and 3). Low-starch, high-degradable AA+ and HSLDAA+ diets were expected to meet the LysDI (7.0% PDI), MetDI (2.4% PDI), and HisDI (2.4% PDI) requirements according to INRA (2018) through 3 RP-AA. We assumed that 100 g AjiPro-L (Ajinomoto Animal Nutrition Europe; Paris, France) would supply 25 g LysDI, while 100 g Smartamine-M provided 60 g MetDI (Adisseo, Commentry, France), and 100 g RP-His (Ajinomoto) provided 20 g of His, based on the manufacturer's specifications. Considering the AA content of AjiPro-L (40% Lys), Smartamine-M (75% Met), and RP-His (45% His), these manufacturer-provided values (i.e., [AA content × bioavailability]) correspond to bioavailability estimates (i.e., the percentage of digestible AA in the product) of 64% for Lys in AjiPro-L G3 (Hultquist et al., 2017) and 80% for Met in Smartamine-M (Whitehouse et al., 2016). For RP-His, the implied bioavailability (45%) was higher than the 38% reported by Whitehouse et al. (2019). Table 1 shows the proportions (% DM) of AjProL, Smartamine, and RP-His introduced into the AA+ diets.

Table 1. Ingredients in the 4 diets[1]

	LSHD		HSLD	
Ingredient (% DM)	AA−	AA+	AA−	AA+
Corn silage	25.1	24.9	47.8	47.0
Grass haylage[2]	38.1	37.7	—	—
Hay[3]	—	—	9.2	9.2
Alfalfa hay	—	—	5.86	5.80
Concentrates[4,5]	34.6[4]	34.2[4]	36.6[5]	36.4[5]
Sugarcane molasses	1.95	1.90		
Wheat bran	—	0.334	—	0.345
Calcium carbonate	—	0.217	—	0.259
Smartamine-M[6]	—	0.053	—	0.074
AjiPRo-L[7]	—	0.304	—	0.267
RP-His[7]	—	0.145	—	0.115
Urea	0.250	0.247	0.540	0.540

[1]LSHD: low-starch, high-degradable in the rumen; HSLD: high-starch, low-degradable in the rumen; AA−: without AA supplementation; AA+: with rumen-protected AA supplies (Lys, Met, and His).

[2]Italian ryegrass haylage.

[3]Permanent grassland, lowlands.

[4]Low-starch, high-degradable in the rumen concentrate (Garun-Paysanne, Montauban de Bretagne, France) contained (as % DM): 20.6% wheat, 20.0% barley, 7.3% soybean meal, 3.1% rapeseed meal (00: without erucidic acid and low glucosinolates), 5.8% corn gluten meal, 32.8% dried beet pulp, 2% wheat bran, 3% palm oil, 1.7% sugarcane molasses, 0.95% calcium carbonate, 0.1% NaCl, 1% sodium bicarbonate, 0.65% calcium phosphate, 0.2% magnesium oxide, and 0.8% mineral and vitamin premix. The mineral and vitamin premix (Provimi France, SAS, Crevin, France) was composed of Ca: 36%, Mg: 0.2%, vitamin A: 2,000,000 IU/kg, vitamin D_3: 500,000 IU/kg, vitamin E: 3,750 mg/kg, vitamin B_1: 1,000 mg/kg, Cu: 5,000 mg/kg, Zn: 17,500 mg/kg, Mn: 12,500 mg/kg, I_2: 325 mg/kg, Co: 90 mg/kg, and Se: 100 mg/kg).

[5]High-starch, low-degradable in the rumen concentrate contained (as % DM): 52.3% ground corn, 22.0% soybean meal, 3.5% rapeseed meal (00: without erucidic acid and low glucosinolates), 9.9% dehydrated alfalfa, 3.5% corn gluten meal, 1% wheat bran, 2.4% palm oil, 1.7% sugarcane molasses, 0.95% calcium carbonate, 0.1% NaCl, 1% sodium bicarbonate, 0.65% calcium phosphate, 0.2% magnesium oxide, and 0.8% the same mineral and vitamin premix as the LSHD concentrate.

[6]Rumen-protected Met (Adisseo, Commentry, France).

[7]Rumen-protected Lys (AjiPRo-L) and His from Ajinomoto Animal Nutrition Europe (Paris, France).

Feeding, Measurements, Sample Collection, and Preparation

Feeding. The amount of feed offered and the refusals of individual cows were recorded daily during the experiment. Two different corn silages had to be used sequentially to feed all cows throughout the experiment. The second corn silage started to be distributed January 19, 2021 (cows were at 108 ± 18 DIM). In contrast, a single Italian ryegrass haylage, harvested from one field, was used throughout the entire experimental period. A small sample of each wrapped round ball of haylage (50 g fresh weight) was taken using a forage coring auger. These samples were analyzed before haylage distribution using the near-infrared spectroscopy method (DS2500, FOSS, Nanterre, France) at the Cargill Laboratory (Provilab, Yffiniac, France) to record any change in the characteristics of the feed, and particularly in CP. Based on these CP analyses, 3 balls of haylage were then mixed to be distributed together to limit variations in the CP content (estimated by NIRS as 127.0 ± 4.7 g/kg DM). Two samples (2 × 800 g fresh weight) of distributed haylage were collected 3 times a week, and corn silage samples (2 × 800 g fresh weight) were collected 5 times a week. In addition, other forages (2 × 500 g fresh weight) and concentrate samples (2 × 200 g fresh weight) were collected once a week. One sample of each feed ingredient was dried for 48 h at 60°C in a forced-air oven to determine the DM content and for analyses. The other samples were frozen at −20°C. Each type of sample (dried or frozen) was pooled differently depending on the analyses performed. First, dried samples from the first and second corn silages were each grouped into 2 pools, each representing half of their respective periods of use (corn silage 1: November 17 to December 18, 2020, and December 21, 2020, to January 18, 2021; corn silage 2: January 19 to March 11, 2021, and March 12 to May 2, 2021). Grass haylage dried samples were pooled into 4 groups following the same pattern as the corn silages. Each of these dried samples was then ground through a 0.8-mm grid for chemical analyses (DM, Ash, CP, NDF, ADF, ADL, Crude Fiber, EE, starch, Ca, and P) determined as recommended by INRA (2018) and 4-mm sieves for enzymatic starch analysis only (INRA, 2018). The gross energy (**GE**) content of these samples (ISO, 1998) was measured using an adiabatic bomb calorimeter (IKA C6000, IKA,

Table 2. Chemical and nutrient compositions of the main feedstuffs measured (mean ± SD; g/kg DM, except as noted)

	Corn silage[1]		Grass haylage[2]					
Composition	1	2	1	2	Hay[3]	Alfalfa hay	LSHD conc.[4]	HSLD conc.[5]
Ash	44 ± 0.1	43 ± 2	131 ± 0.2	135 ± 0.4	70	76	81	75
CP	72 ± 3	69 ± 1	124 ± 0.3	126 ± 6	71	118	186	216
Starch	208 ± 1	238 ± 9	—	—	—	—	285	352
Crude fiber	248 ± 6	223 ± 1	254 ± 1	249 ± 3	328	400	83	59
NDF	499 ± 11	456 ± 3	487 ± 2	482 ± 2	661	612	237	151
ADF	280 ± 9	251 ± 1	280 ± 3	284 ± 0.4	372	442	124	75
ADL	23 ± 3	18 ± 3	22 ± 3	25 ± 0.2	47	93	20	19
EE[6]	29 ± 1	27.5 ± 0.1			21	14	65	51
GE[7] (Mcal/kg DM)	4.57	4.54	4.39	4.37	4.30	4.36	4.31	4.41
ED6_N[8] (%)	62.0	69.6	76.6		43.1	67.4	59.2	56.8
ED6_st[9] (%)	89.5	94.8	—	—	—	—	90.6	66.8
AA (g/100 g)								
Lys	0.227	0.223	0.465		0.270	0.581	0.786	0.961
His	0.122	0.085	0.156		0.092	0.209	0.453	0.490
Arg	0.155	0.109	0.305		0.264	0.381	0.861	1.109
Thr	0.267	0.256	0.510		0.291	0.500	0.755	0.875
Val	0.344	0.328	0.633		0.323	0.536	0.875	0.959
Met	0.126	0.133	0.178		0.118	0.164	0.378	0.382
Ile	0.256	0.239	0.500		0.236	0.421	0.712	0.832
Leu	0.602	0.539	0.815		0.454	0.725	1.676	1.860
Phe	0.265	0.217	0.509		0.291	0.478	0.915	1.010
Asp	0.489	0.464	0.905		0.544	1.629	1.428	1.895
Ser	0.274	0.259	0.428		0.276	0.512	0.897	0.960
Glu	0.860	0.764	0.708		0.644	0.970	3.715	3.568
Pro	0.430	0.374	0.629		0.322	0.661	1.336	1.265
Gly	0.316	0.310	0.529		0.329	0.495	0.718	0.840
Ala	0.491	0.502	0.756		0.373	0.506	0.947	1.084
Tyr	0.175	0.148	0.345		0.197	0.392	0.808	0.795
Cys	0.094	0.084	0.105		0.074	0.123	0.344	0.351
Trp	0.201	0.042	0.122		0.105	0.175	0.040	0.243

[1]Two different corn silages were analyzed twice. The 2 different corn silages had to be used sequentially to feed all cows throughout the experiment, with a change started January 19, 2020. The samples of the first and second corn silages were each grouped into 2 pools for chemical composition analysis, with each pool representing half of their respective periods of use (corn silage 1: November 17 to December 18, 2020, and December 21, 2020 to January 18, 2021; corn silage 2: January 19 to March 11, 2021, and March 12 to May 2, 2021).

[2]The grass haylage was analyzed using the same protocol as the 2 corn silages.

[3]Permanent grassland, lowlands.

[4]LSHD concentrate: composition detailed in Table 1.

[5]HSLD concentrate: composition detailed in Table 1.

[6]Ether extract.

[7]Gross energy measured using a bomb calorimeter (IKA C6000).

[8]Effective degradability assuming passage rate of 6%/h, of nitrogen (in situ).

[9]Effective degradability assuming passage rate of 6%/h, of starch (in situ).

Staufen, Germany). Frozen samples of each corn silage and haylage were grouped in one pool. These pooled frozen samples of corn silage and haylage were used to measure fermentable products (INRA, 2018), and the rest was freeze-dried. Second, dried samples of each concentrate, alfalfa hay, and permanent grassland hay collected throughout the experimental period were pooled. Freeze-dried pooled samples of corn silage and grass haylage and dried pooled samples of other feed ingredients were used to assess the in situ ruminal degradation of nitrogen and starch and AA analysis. These samples, ground through a 0.8-mm or 4-mm grid, respectively, were used to determine the effective degradability coefficients for N (ED6_N, %) and starch (ED6_st, %), calculated for a passage rate of 6% per hour. Nylon bags were introduced into the rumen of 3 dried cows according to INRA (2018) and based on Ørskov and MacDonald (1979). For AA analyses, the pooled samples were ground through a 0.5-mm sieve. The AA contents were measured using HPLC chromatography (Alliance System, Waters, Guyancourt, France) after protein hydrolysis with 6 N HCl at 110°C for 23 h under reflux (Directive 98/64/CE, 3/09/99-Norme NF EN ISO 13903; ISO, 2005). The Cys and Met contents were determined after oxidation with performic acid before hydrolysis. The Trp content was determined after hydrolysis at 120°C for 16 h with barium hydroxide and separation by reverse-phase HPLC and fluorometric detection.

Table 3. Energy, MP, and AA supplied by the 4 diets[1]

	LSHD		HSLD	
Item	AA−	AA+	AA−	AA+
Composition (g/kg DM)				
Ash	92.2	94.0	58.3	60.9
CP	138	138	142	143
Starch	156	155	235	233
NDF	384	381	379	377
ADF	216	214	212	210
Crude fiber	182	180	193	192
Ether extract	43	43	35	35
ME (Mcal/kg DM)	2.42	2.39	2.32	2.29
NEL (Mcal/kg DM)	1.52	1.51	1.44	1.42
Starch ED[2] (%)	83.7	83.7	74.1	73.7
Bypass starch[3] (g/kg DM)	25.5	25.3	60.8	61.3
FOM[4] (g/kg DM)	459	455	402	394
MP[5] (g/kg DM of PDI)	88.7	89.3	93.4	94.3
Microbial MP[6] (g/kg DM of PDIM)	49.8	49.4	47.2	46.6
Digestible RUP[7] (g/kg DM of PDIA)	38.9	40.0	46.2	47.7
Ruminal protein balance[8] (g/kg DM)	−3.0	−2.9	−2.7	−2.7
AADI[9] (% PDI)				
LysDI	6.54	7.30	6.49	7.10
MetDI	2.15	2.48	2.09	2.54
HisDI	2.07	2.37	2.11	2.33
ArgDI	4.58	4.51	4.73	4.67
IleDI,	5.02	4.94	4.87	4.79
LeuDI	8.95	8.80	9.06	8.93
PheDI	5.17	5.09	5.09	5.02
ThrDI	5.14	5.06	5.08	5.00
ValDI	5.63	5.54	5.47	5.39
AlaDI	6.75	6.65	6.67	6.57
AspDI	10.13	9.98	10.36	10.21
GluDI	14.70	14.48	14.77	14.58
GlyDI	6.42	6.33	6.38	6.28
ProDI	5.18	5.10	5.01	4.95
SerDI	4.81	4.73	4.81	4.74
TyrDI	4.15	4.09	4.09	4.03

[1]LSHD: low-starch, high-degradable in the rumen; HSLD: high-starch, low-degradable in the rumen; AA−: without AA supplementation; AA+: with rumen-protected AA supplies (Lys, Met, and His).

[2]Effective degradability of starch for each diet estimated by INRA (2018) using the effective degradability of feedstuffs (Table 1).

[3]Starch_duodenum (g/kg DM) = starch (g/kg DM) × (1 − 0.01 × starch ED).

[4]Fermentable OM calculated using INRA (2018).

[5]PDI: true protein digested in the small intestine equivalent to MP in INRA (2018).

[6]PDIM: true protein digested in the small intestine from microbes, that is, digestible microbial MP (INRA, 2018).

[7]PDIA: true protein digested in the small intestine from alimentary (feedstuffs), that is, digestible RUP (INRA, 2018).

[8]Estimated by the difference between CP intake and the sum of RUP + microbial CP + endogenous CP. This calculation reflects the equilibrium between fermentable OM and nitrogen supply for microbial synthesis. As endogenous CP entering in the duodenum is set at 14.2 g/kg DM, a negative rumen protein balance (≤14.2 g/kg DM) corresponds to microbial protein synthesis limited by degradable CP. (INRA, 2018).

[9]Amino acids digestible in the intestine (INRA, 2018).

Milk Yield and Milk Composition, BW, and BCS. Individual milk yields were recorded at each milking, from calving until the end of the treatment. Milk fat and true protein contents were measured on 2 d/wk (Tuesday and Thursday) during the morning and afternoon milkings, and the milk lactose and urea contents were measured once a week (on Tuesday), at each milking. Milk fat, protein, urea, and lactose contents were measured using mid infrared analysis (MilkoScan, Foss Electric, Hillerod, Denmark). In addition, morning and evening milk samples were collected 3 times during the trial (68 ± 4, 122 ± 4, and 171 ± 3 DIM) and pooled per day based on the morning and afternoon milk yields. These 3 milk samples were used to determine the contents in casein, milk urea, milk fatty acids, and GE. Milk N, NPN, and noncasein N (**NCN**) fractions were analyzed using the Kjeldahl method, and CN content corresponded to N minus NCN. Milk urea was analyzed as described by Haque et al. (2015). The GE content of milk was measured with a bomb calorimeter (IKA C6000) on freeze-dried milk samples. The ratio between the *trans*-10 C18:1 and *trans*-11 C18:1 milk fatty acids as an indicator of acidosis was reported after measuring the fatty acid profile by the GC method (HP7890A; Agilent Technologies, Diegem, Belgium) after extracting total lipids from milk and methylation according to the methods described by Jing et al. (2018). The cows were weighed each morning at the end of milking on the in-line scale coming out of the parlor. Body condition score was determined by 2 different people once a month using a 1 to 5 scale.

Rumen and Blood Sample Collection. Rumen fluid was collected twice by rumen intubation on the 44 cows at 131 ± 4 DIM, at 1100 h, 3 h after feeding, and at 142 ± 4 DIM at 1400 h, 6 h after feeding. Rumen fluid samples were strained through 6 layers of cheesecloth, and 8 mL of strained rumen fluid was kept at −20°C after mixing with 0.8 mL of 5% (vol/vol) orthophosphoric acid containing 1% (wt/vol) mercury chloride. Volatile fatty acids were determined by GC (Agilent Technologies 7890A GC System, Les Ulys, France; with an SGE column BP20, 15 m length, 0.5 mm diameter, Trajan Scientific, Victoria, Australia) using 4-methylvaleric acid as the internal standard. The molar ratio between rumen propionate and acetate was then calculated. Blood was collected monthly until the end of experiment from the caudal vein, at least 30 min after morning milking and 30 min before feed access (at 73 ± 4, 105 ± 4, 121 ± 6, 150 ± 6, and 176 ± 4 DIM). Blood samples were collected using evacuated EDTA tubes (Vacutest Kima, Azergrande, Italy) for insulin aliquots and evacuated sodium heparin tubes (Vacutest Kima) for glucose, BHB, urea, triglycerides, acetate, lactate, and AA aliquots. Blood plasma was separated by centrifugation at 2,000 × *g* for 15 min at 4°C. Acetate and lactate aliquots were deproteinized with perchloric acid and filtered. All aliquots were stored at −20°C, except those used for AA, which were stored at −80°C. Plasma glucose, BHB, urea, triglycerides, nonesterified fatty acids (**NEFA**), acetate, and lactate were measured using a Konelab (Thermo Scientific, Cergy Pontoise, France) multiparameter analyzer with the following enzymatic kits (Thermo Scientific, Cergy Pontoise, France): glucose with glucose hexokinase; BHB with BHB dehydrogenase; triglyceride with lipase, glycerol kinase, glycerol-3-phosphate oxidase, and peroxidase; NEFA with acetyl-CoA synthase, acyl-CoA oxidase, and peroxidase; urea with urease and glutamate dehydrogenase; lactate with lactate oxidase and peroxidase; and acetate with acetate acetyl-CoA synthase and malate dehydrogenase. Plasma insulin was measured using an immunoenzymatic method (TOSOH Europe, Tessenderlo, Belgium). Plasma AA were measured using an ultra-performance liquid chromatography-MS system (Waters Acquity UPLC System; Waters Corp.) equipped with an Acquity Tunable UV Detector and a mass detector (Simple Quadrupole Detector) for the few coeluting peaks.

Mammary Biopsies. To measure mammary cell proliferation and apoptosis, 2 mammary tissue samples were collected by several biopsies (1 incision with 1 to 3 punctures) from multiparous cows only (between 2 h and 4 h after the morning milking), the first time after 51 d of treatment (105 ± 4 DIM) and second at the end of the experiment (after 124 d of treatment, i.e., 178 ± 3 DIM). These mammary biopsy samples were taken from the upper portion of the mammary gland, alternating between the right and left rear glands, at the 2 sampling periods (51 d and 124 d) as described by Herve et al. (2019). The evening before the biopsy, the mammary glands were shaved, and the cows received antibiotic prophylaxis consisting of an i.m. injection of 60 mL of Shotapen (Virbac, Carros, France) containing benzylpenicillin (130.38 mg/mL) and dihydrostreptomycin (164.00 mg/mL; Shotapen, Virbac, Carros, France) and 20 mL i.v. Metacam (Boehringer Ingelheim, Ingelheim am Rhein, Germany) containing 20 mg/mL meloxicam (a nonsteroidal anti-inflammatory drug). The biopsy sites were cleaned with alcohol and povidone-iodine (Vétédine Savon, Vétoquinol, Magny-Vernois, France) and locally anesthetized with procaine (Procamidor 20 mg/mL, Axience, Pantin, France). The biopsy site skin was incised with a scalpel. A biopsy tool was set up on a drill as described by Farr et al. (1996). The skin incision was closed with a disposable skin stapler (Royal 35W, Clinique Veterinaire, Saint-Gregoire, France) and protected with an aluminum suspension dressing (Aluspray, Vétoquinol, Magny-vernois, France). Immediately after the procedure was completed, the cows were milked manually to remove any blood from the mammary gland. The tissue specimens were prepared as described by Herve et al. (2019). Cell proliferations were measured by immunohistochemical staining using the proliferation cell nuclear antigen (**PCNA**) method. Determination of the percentage of apoptotic cells in the biopsy sections was based on DNA fragmentation detection using terminal deoxynucleotidyl transferase–mediated 2′-deoxyuridine 5′-triphosphate nick-end labeling. The 2 methods were described by Herve et al. (2019).

Data Processing and Analysis

Data processing and statistical analysis were performed using R (R.4.3.1, R Core Team, 2020) with the IDE RStudio (Posit Team, 2025).

Dry matter intake, milk yield, MPY, MFY, lactose yield, MPC, MFC, lactose content, and BW records were checked for outliers according to the procedure used by Friggens et al. (1999). Briefly, for each parameter and each cow, cubic spline curves were fitted using the smooth.spline function from the STAT package (R.4.3.1, R Core Team, 2020), and any observed data seen to deviate from the spline by ±5 × SD of the model were excluded. This model was run twice to eliminate extremely deviant points. The procedure resulted in the elimination of very few points on the initial dataset. (e.g., for DMI: 25 points eliminated out of 7,885 points recorded; milk yield: 55 points eliminated out of 7,845 points recorded). The method was described in the Supplemental R Scripts (see Notes).

Using this outlier-free dataset, and to determine covariates, we first fitted a polynomial curve from 1 to 49 DIM with the lme function from the nlme package (Pinheiro and Bates, 2000) according to the following model:

$$Y_{ijk} = \mu + \text{Parity}_i + \beta_j \times \text{DIM} + (\text{Cow}_j + \beta_j \times \text{DIM}) + \varepsilon_{ijk},$$

where Y_{ijk} is the dependent variable for the fixed effects of Parity_j (primiparous or multiparous), μ is the intercept, DIM is fitted as a continuous variable (1–49 DIM), β_j are the slopes (Cow_j × DIM) and their interactions, and ε_{ijk} is the residual error associated with ijk observations. Cow was treated as a random effect on the intercept, and the slope assumed to be ~iid$N(0, \sigma k^2)$. Polynomial curves from 1 to 4 degrees were tested, and degree 4 was finally chosen because of its better adjustment according to Akaike information criterion. In a second step, fitted values from 21 to 35 DIM were averaged to determine the covariate. This 2-step procedure allowed us to determine covariates for the main analysis that were well-adjusted due to the elimination of extreme values causing polynomial regression end effects.

Finally, during the treatment period (56–183 DIM), the data were analyzed using the following regression model with the same package as that used for covariates:

$$Y_{ijklm} = \mu + \text{CovY}_i + \text{Block}_j + \text{Starch}_k + \text{AA}_l + \text{Starch}_k \times \text{AA}_l + \beta_j \times \text{DIM} + \beta_k \times \text{DIM} + \beta_l \text{Block} \times \text{DIM} + \beta_{kl} \times \text{DIM} + \text{Cow}_m + \beta_m \times \text{DIM} + \varepsilon_{ijklm},$$

where μ is the intercept; Y_{ijklm} is the dependent variable for the fixed effects of covariate_i (CovY_i, 44 cows); Block_j (11 blocks); Starch_k (LSHD vs. HSLD); AA_l (AA− vs. AA+); DIM is fitted as a continuous variable (56 to 183 DIM); and their interactions with their slopes β_j (Block × DIM); β_k (starch × DIM); β_l (AA × DIM); β_{kl} (starch × AA × DIM); β_m (Cow × DIM); and ε_{ijklm} is the residual error associated with ijklm observations. Cow was treated as a random effect on the intercept and the slope. Milk yield, milk composition, and DMI were adjusted as indicated in the covariates model. Day in milk was centered on d 118 to minimize correlations between slope and intercept. We assumed that the slopes, calculated as a combination of β_i by the Emtrends function of R (package emmeans) according to each treatment, represented persistency for milk and milk components and ADG.

Plasma metabolites, AA, insulin, milk fatty acid profile, and concentrations were analyzed using the same package according to the following model:

$$Y_{ijklm} = \mu + \text{CovY}_i + \text{Block}_j + \text{Starch}_k + \text{AA}_l + \text{Time}_m + \text{Starch}_k \times \text{AA}_l + \text{Starch}_k \times \text{Time} + \text{AA}_l \times \text{Time} + \text{Block}_j \times \text{Time} + \text{Starch}_k \times \text{AA}_l \times \text{Time} + \text{Cow}_i + \varepsilon_{ijklm},$$

where μ is the mean; Y_{ijklm} is the dependent variable for the fixed effects of covariate_i (CovY_i, 44 cows); Block_j (11 blocks of 4 cows with 3 blocks of primiparous); Starch_k (LSHD vs. HSLD); AA_l (AA− vs. AA+); Time_m (1 to 5 or 1 to 3), and their interactions; and ε_{ijklm} is the residual error associated with ijklm observations. Cow was treated as a random effect. The autocorrelation structure of order 1 was used as the covariance structure. Data transformation with a natural logarithm was applied to BHB, NEFA, lactate, and insulin concentrations. For each ANOVA, any data that deviated from ±3 standardized residues of the model were excluded.

Cell proliferation and apoptosis were analyzed after log-transformation and quasibinomial distribution due to over-dispersion on the counted data, using the same model as that described above but with the glmmPQL function from the MASS package (Venables and Ripley, 2002).

The level of significance was set at $P \leq 0.05$ and the trend at $0.05 < P \leq 0.10$. The normality of residues was checked graphically with quantile-quantile plot. When an interaction was significant or tended to be significant, the interaction effect was tested by one-way ANOVA for each factor level (contrast test).

RESULTS

Diets, DMI, ME, and Protein and AA Intakes

The starch content was +78.5 g/kg DM higher in the 2 HSLD diets than in the 2 LSHD diets (Table 3), and its effective rumen degradability decreased in HSLD versus LSHD diets as expected (Table 3), leading to a higher bypass starch content (+35.7 g/kg DM) in HSLD versus LSHD diets. Metabolizable energy and NEL contents were slightly lower in HSLD versus LSHD diets (Table 3), which was not intended. Crude protein contents were close to 140 g/kg DM but higher in HSLD than in LSHD

Table 4. Effects of increasing the bypass starch content and AA supplementation[1] on nutrient intakes in dairy cows from 56 to 183 DIM

	LSHD		HSLD			*P*-value[2]						
Item	AA−	AA+	AA−	AA+	SEM	St.	AA	St. × AA	T	St. × T	AA × T	St. × AA × T
DMI (kg/d)	23.8	24.1	23.9	24.7	0.35	0.25	0.11	0.52	<0.01	<0.01	0.22	0.10
DMI slope (g/d)	−3.9	3.9	17.3	21.2	4.43							
ME intake (Mcal/d)	57.5	57.9	55.5	56.7	0.72	0.03	0.23	0.63	<0.01	<0.01	0.18	0.61
NEL intake (Mcal/d)	36.3	36.4	34.5	35.2	0.44	<0.01	0.29	0.65	<0.01	<0.01	0.16	0.62
CP intake (g/d)	3,286	3,349	3,414	3,554	50.3	<0.01	0.05	0.45	<0.01	0.07	0.22	0.67
MP intake[3] (g of PDI/d)	2,113	2,163	2,240	2,344	35.3	<0.01	0.04	0.45	<0.01	<0.01	0.28	0.60
Starch intake (g/d)	3,730	3,768	5,636	5,798	91.7	<0.01	0.28	0.50	<0.01	<0.01	0.11	0.49
Bypass starch (g/d)	610	617	1,464	1,529	35.8	<0.01	0.32	0.42	<0.01	<0.01	0.26	0.45
Intestinal glucose[4] (g/d)	432	437	974	1,017	22.6	<0.01	0.30	0.41	0.01	<0.01	0.26	0.47
Acetate[4] (g/d)	3,391	3,409	2,955	2,972	33.5	<0.01	0.61	0.99	<0.01	<0.01	0.20	0.71
Propionate[4] (g/d)	1,649	1,658	1,470	1,509	23.8	<0.01	0.33	0.55	<0.01	<0.01	0.19	0.69
Butyrate[4] (g/d)	978	983	876	886	10.3	<0.01	0.46	0.83	<0.01	<0.01	0.12	0.70
LysDI[5] (g/d)	139	158	145	167	2.5	<0.01	<0.01	0.76	<0.01	<0.01	0.36	0.53
MetDI[5] (g/d)	45.5	53.6	46.8	59.5	0.87	<0.01	<0.01	0.01	0.01	<0.01	0.22	0.38
HisDI[5] (g/d)	43.8	51.2	47.4	54.7	0.79	<0.01	<0.01	0.96	<0.01	<0.01	0.25	0.63
ILeDI[5] (g/d)	106	107	109	112	1.66	0.01	0.24	0.47	<0.01	<0.01	0.30	0.65
LeuDI[5] (g/d)	189	190	203	209	3.22	<0.01	0.23	0.44	<0.01	<0.01	0.27	0.61
ValDI[5] (g/d)	119	120	122	126	1.91	0.01	0.21	0.48	<0.01	<0.01	0.30	0.76

[1]LSHD: low-starch, high-degradable in the rumen; HSLD: high-starch, low-degradable in the rumen; AA−: without AA supplementation; AA+: with rumen-protected AA supplies (Lys, Met, and His).

[2]Probability values correspond to the effect of the starch content (St.: low starch vs. high starch); the effect of AA supplementation (AA: AA− vs. AA+); the effect of time (T); and interactions between starch, AA, and time: St. × AA, St. × T, AA × T, St. × AA × T. When interactions between treatments × time are significant, the next line shows the corresponding slopes of the variable.

[3]PDI: true protein digested in the small intestine supplied by digestible RUP and digestible rumen microbial protein (INRA, 2018), equivalent to MP.

[4]Available intestinal glucose = DMI × starch_duodenum × 0.01 × Std_si (Starch_digestibility in small intestine), with starch_duodenum (g/kg DM) = starch (g/kg DM) × (1 − 0.01 × starch ED) with starch ED is the effective degradability of starch for each diet (Table 3) and Std_si (%) = 74.05 − 0.122 × starch_duodenum (g/kg DM)] as predicted in INRA (2018); ruminal acetate (g/d) = DMI × TVFA/MW × [54.2 + 12 × log10 (100 × DNDF/DOM) − 0.052 × Starch ED − 1.99 × (DMI × 100)/BW], with TVFA (Total VFA, mol/kg DM) = [8.35 − 1.1 × (PCO − 0.43)] × 0.001 × FOM (g/kg DM), PCO: the proportion of concentrate (<1; Table 1), FOM: fermentable OM (Table 3), MW: the molecular weight of acetate, DNDF/DOM (g/g): the ratio between digestible neutral fiber and digestible OM; ruminal propionate (g/d) = DMI × TVFA/MW × [19.7 − 6.63 × log10 (100 × DNDF/DOM) + 0.070 × Starch ED + 2.62 × (DMI × 100)/BW]; and butyrate (g/d) = DMI × TVFA/MW × [19 × −3.99 × log10 (100 × DNDF/DOM) − 0.026 × Starch ED], as predicted in INRA (2018).

[5]Lys, Met, His, Leu, and Val digestible in the intestine (AADI; INRA, 2018) in g/d.

diets (Table 3), and MP contents were also higher in HSLD versus LSHD diets because of more digestible RUP (Table 3), which was unexpected. As expected, the microbial MP content (PDIM: protein digestible in the intestine from microbes; INRA, 2018) was slightly lower in the HSLD diets than in the LSHD diets. However, the rumen protein balance was close to zero (Table 3), as intended, due to a higher urea supply in the HSLD diets versus the LSHD diets (Table 1). The AA contents in the AA+ diets (LysDI = 7.2% of PDI; MetDI = 2.5% of PDI; HisDI = 2.4% of PDI) were increased to be close to INRA (2018) requirements (LysDI = 7.0% of PDI; MetDI = 2.4% of PDI; and HisDI = 2.4% of PDI) versus in AA− diets (LysDI = 6.5% of PDI; MetDI = 2.1% of PDI; and HisDI = 2.1% of PDI). Neutral detergent fiber contents were similar in the diets, as expected (Table 3).

Mean DMI did not vary among treatments ($P \geq 0.11$; Table 4; Figure 1). Daily DMI did not change over time with the 2 LSHD diets (mean of the LSHDAA− and LSHDAA+ slopes = 0 ([−3.9+3.9]/2, Table 4) but increased over time with HSLD diets (mean of the LSHDAA− and LSHDAA+ slopes = +19.3 g/d); starch × time interaction: $P < 0.01$). At the end of the experiment (183 DIM), the HSLD group consumed 1.57 kg/d of DM more than the LSHD group [ΔMeanDMI + Δslopes/1,000 × 127/2 d = (24.30 − 23.95) × (19.2 − 0)/1,000 × 127/2 = 1.57]. In addition, a tendency toward a starch × AA × time interaction ($P = 0.10$) was observed on DMI, indicating that the slope of DMI tended (based on the contrast test: $P = 0.10$) to be lower in LSHDAA− (−3.9 g/d) than in LSHDAA+ (+3.9 g/d).

Metabolizable energy (−1.6 Mcal/d, $P = 0.03$) and NEL intakes (−1.5 Mcal/d, $P < 0.01$) were lower in HSLD than in LSHD diets (Table 4). The starch intake (+1,968 g/d, $P < 0.01$) and bypass starch intake (+883 g/d) were higher in HSLD diets than in LSHD diets. Predicted intestinal glucose levels were higher in HSLD diets than in LSHD diets (+561 g/d, $P < 0.01$), but predicted acetate, propionate, and butyrate amounts were higher with LSHD diets than with HSLD diets (+437 g/d, +164 g/d, and +100 g/d, respectively, $P < 0.01$). The molar ratio between rumen propionate and acetate did not differ significantly ($P = 0.29$) between HSLD (3.73 ± 0.08) and LSHD diets (3.81 ± 0.08). In addition, the ratio between *trans*-10 C18:1 and

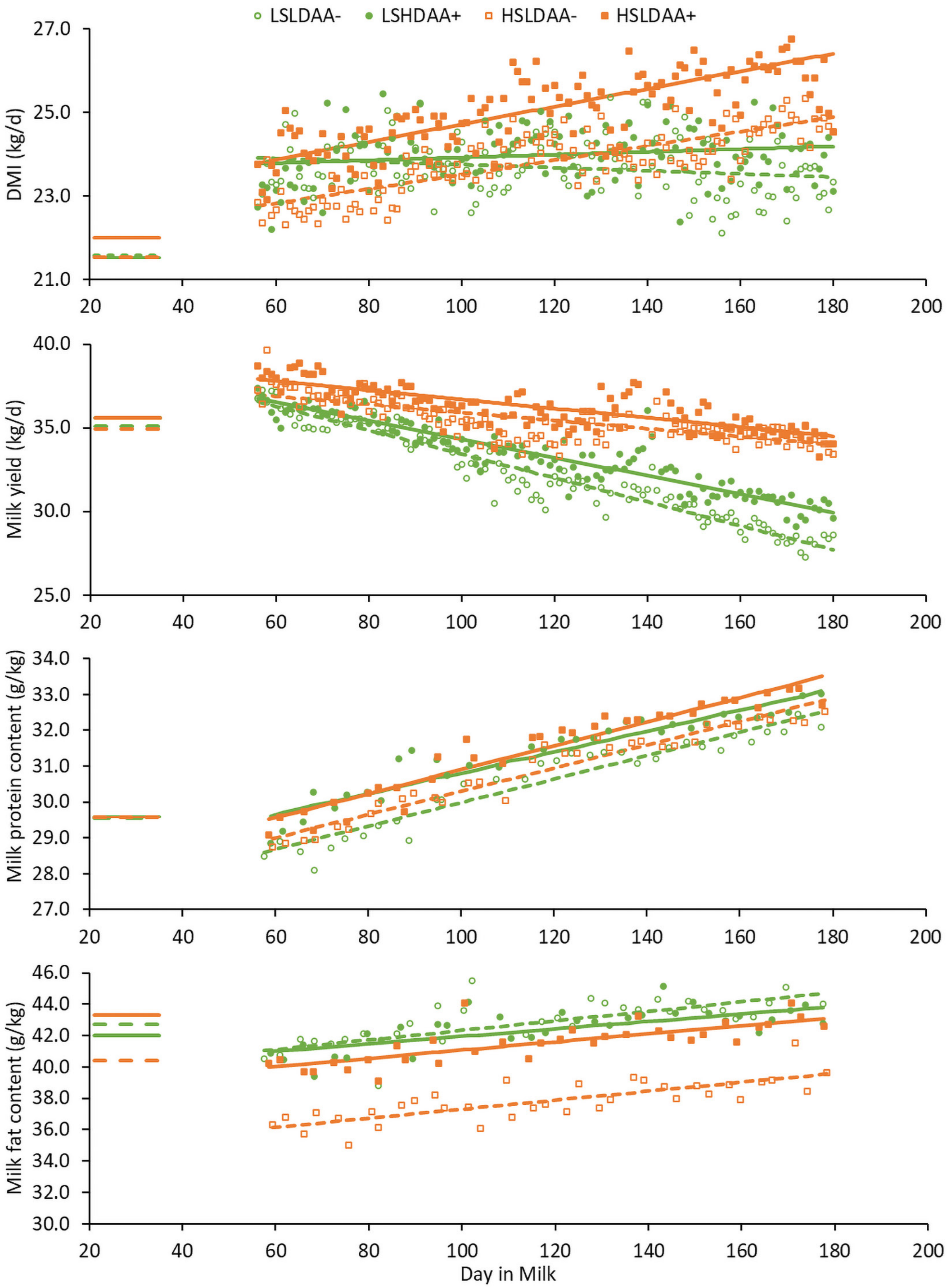

Figure 1. Effects of increasing the bypass starch content and AA supplementation on lactation persistency, milk yield, and true protein and fat contents from 56 to 183 DIM. Symbols represent the means of raw data per dietary treatment: open green circle = LSHDAA−, low-starch high-degradable in the rumen without AA supplementation; solid green circle = LSHDAA+, low-starch high-degradable in the rumen with rumen-protected Lys, Met, and His supplementation; open orange square = HSLDAA−, high-starch low-degradable in the rumen without AA supplementation; solid orange square = HSLDAA+, high-starch low-degradable in the rumen with rumen protected Lys, Met, and His supplementation. Lines between 21 and 35 DIM represent covariables (dashed green line = LSHDAA−; solid green line = LSHDAA+; dashed orange line = HSLDAA−; solid orange line = HSLDAA+). From 56 to 183 DIM, the mean slopes are also represented by covariable lines (dashed green line = LSHDAA−; solid green line = LSHDAA+; dashed orange line = HSLDAA−; solid orange line = HSLDAA+). The mean of the individual slopes is not mathematically the same as the slopes presented in Tables 4 and 5, because the statistical model included the covariable correction. The increases in milk yield around 140 DIM were due to a computer problem in the milking parlor that caused a lack of some data (n ≥7 instead of n = 11 per treatment).

trans-11 C18:1 milk fatty acids did not differ between HSLD (0.565) and LSHD (0.571) diets (SEM = 0.015, P = 0.66), suggesting that there was no acidosis problem.

Metabolizable protein intake was higher under HSLD diets than LSHD diets (Table 4: +154 g/d PDI, $P < 0.01$) and with AA supplementation (+77 g/d PDI, $P < 0.01$). Digestible Lys, Met, and His intake (g/d) were higher with AA supplementation (+20.5 g/d, +10.3 g/d, and +7.4 g/d, respectively, $P < 0.01$; Table 4). Because the MP intake was higher on HSLD versus LSHD diets, digestible Lys, Met, and His supplies were also significantly increased on HSLD versus LSHD diets (+7.5, +3.7, and +3.6 g/d). A starch × AA interaction was observed for MetDI intake (g/d) because MetDI supply was higher in HSDAA+ than in LSDAA+ treatments. Digestible AA intake (g/d), such as branched-chain AA (**BCAA**: Ile, Leu, and Val; shown in Table 4), increased on the HSLD versus LSHD diets due to higher MP intakes. Overall, MP and NEL intake (/d) and consequently all the estimates of nutrient supplies increased with time in HSLD versus LSHD diets (starch × time) because DMI slopes increased in HSLD versus LSHD diets.

Milk Yield and Composition

Bypass Starch Effect. The mean milk yield was higher on HSLD than on LSHD diets (+3.1 kg/d, $P < 0.01$), and lactation persistency (+36.5 ± 5.4 g/d, mean ± SEM, i.e., differences in milk yield slopes in Table 5; Figure 1) was higher on HSLD versus LSHD diets (time × diet: $P < 0.01$). At the end of experiment (183 DIM), the HSLD group produced +5.4 kg/d more milk than the LSHD group (calculated from the slope and mean milk yields). Similar to the milk yield, lactose yield also increased with HSLD versus LSHD diets (1,607 g/d vs. 1,759 g/d, $P < 0.01$; Table 5; Figure 2), the slope being less negative with HSLD versus LSHD diets (starch × time interaction), suggesting a greater persistency with HSLD versus LSHD diets in terms of lactose secretion (difference in HSLD vs. LSHD slopes: +1.8 ± 0.27 g/d of lactose). The mean MPY was higher on HSLD diets than LSHD diets (+101 ± 17.2 g/d, $P < 0.0001$). Milk protein yield curves declined over time on LSHD diets but increased on HSLD diets (slopes: −0.90 g/d vs. +0.40 g/d in LSHD vs. HSLD, respectively, $P < 0.01$), suggesting that HSLD diets also increased MPY persistency. The mean MFY did not differ between HSLD and LSHD (P = 0.2). However, a starch × time interaction was observed, and the MFY slope was less negative on HSLD than on LSHD treatments (Table 5). The MFY slope increased by 1.75 ± 0.27 g/d in HSLD versus LSHD diets. In fact, MFY was lower at 56 DIM (−80 ± 8.8 g/d) and higher at 182 DIM (+138 g/d) on HSLD versus LSHD diets (Figure 2). On average, HSLD slightly decreased the milk fat content (−2.6 g/kg, $P < 0.01$) when compared with LSHD.

No interaction between starch and AA was found in any of the milk yield or composition parameters (Table 5: P-value for starch [St.] × AA). In addition, the effects of the treatments on milk lactose, protein, or fat contents were independent of time (Figure 1 and Table 5). Indeed, no time × starch or time × AA interactions on these milk contents were significant.

Rumen-Protected AA Effects. Rumen-protected AA supplementation did not affect milk yield and persistency except with the LSHDAA− diet (starch × AA × time: P = 0.06; Table 5), which tended to have the lowest persistency (contrast test: LSHDAA− vs. LSHDAA+: −72 g/d vs. −54 g/d, P = 0.04) of the 4 treatments. This resulted in +1.70 kg of milk produced in LSHDAA+ versus LSHDAA− at 183 DIM (Table 5; Figure 1). A significant starch × AA × time interaction was also observed on the lactose yield curve (P = 0.01), showing that LSHDAA− presented lower lactose yield persistency than LSHDAA+ (contrast test: P = 0.03). True milk protein content tended to increase with AA+ versus AA− diets (31.4 vs. 30.8 g/kg, P = 0.07), with no AA × time interaction. Milk protein yield also increased on AA+ versus AA− diets (+37 g/d, P = 0.04), with a tendency ($P < 0.10$) for a time × starch × AA interaction, but the contrast test for the slope of MPY between LSHDAA+ versus LSHDAA− was not significant (P = 0.15). Milk fat yield tended to increase (+ 44 g/d in AA+ vs. AA− diets, P = 0.06), with a starch × AA × time interaction (P = 0.02). The contrast test (P = 0.096) indicated that the HSLDAA+ slope for MFY (−0.3 g/d) tended to be lower than that for HSLDAA− (+0.2 g/d). However, at the end of this experiment (183 DIM), the difference in MFY between HSLDAA+ versus HSLDAA− was ~+46 g/d.

BW and Efficiencies

Mean BW did not differ among the treatments ($P > 0.5$), but it varied over time and tended to increase more over time (slope: +104 g/d, P = 0.07, Table 5) with HSLD diets than LSHD diets, leading to a difference at the end of the experiment of +14.9 kg (slope × 127 d). Body condition score did not vary among treatments but varied with time and time × treatment as lactation progressed.

Feed efficiency (milk yield/DMI, Table 5) increased with HSLD diets compared with LSHD diets (+0.09 kg/kg, $P < 0.01$). Milk energy exported in milk per day (Table 5) was higher with HSLD diets than with LSHD diets (+1.45 Mcal/d, $P < 0.01$) and tended to increase with AA supplementation (+0.75 Mcal/d, P = 0.06). The ME efficiency for milk (i.e., the ratio between energy

Table 5. Effects of increasing the bypass starch content and AA supplementation[1] on performance

	LSHD		HSLD			P-value[2]						
Item	AA−	AA+	AA−	AA+	SEM	St.	AA	St. × AA	T	St. × T	AA × T	St. × AA × T
Milk yield (kg/d)	32.4	33.2	35.7	36.1	0.54	<0.01	0.30	0.67	<0.01	<0.01	0.18	0.06
Milk slope (g/d)	−72	−54	−25	−28	5.4							
Lactose content (g/kg)	49.2	48.9	49.2	49.2	0.36	0.64	0.71	0.74	<0.01	0.41	0.35	0.53
Lactose yield (g/d)	1,593	1,620	1,753	1,765	30.1	<0.01	0.51	0.78	<0.01	<0.01	0.20	0.01
Lactose slope	−3.3	−2.3	−0.9	−1.2	0.27							
True protein content (g/kg)	30.6	31.3	30.9	31.5	0.37	0.54	0.07	0.85	<0.01	0.48	0.67	0.45
Casein[3] (% TP)	81.3	82.2	81.5	80.4	0.46	0.44	0.42	0.28	0.88	0.80	0.96	0.15
True protein yield (g/d)	990	1,033	1,097	1,128	17.2	<0.01	0.04	0.71	<0.01	<0.01	0.48	0.098
Protein slope (g/d)	−1.1	−0.7	0.5	0.3	0.17							
Fat content (g/kg)	42.5	42.5	39.2	40.6	0.69	<0.01	0.30	0.30	<0.01	0.97	0.39	0.73
Fat yield (g/d)	1,386	1,395	1,380	1,459	22.9	0.20	0.06	0.14	<0.01	<0.01	0.74	0.02
Fat slope (g/d)	−2.1	−1.5	0.2	−0.3	0.27							
BW (kg)	643	645	650	644	4.6	0.50	0.65	0.43	<0.01	0.07	0.60	0.61
BW change (g/d)	176	115	249	249	58.0							
BCS	2.03	2.05	2.02	2.07	0.051	0.88	0.52	0.75	<0.01	<0.01	0.05	0.59
Milk yield/DMI (kg/kg DM)	1.37	1.40	1.49	1.45	0.030	<0.01	0.76	0.24	<0.01	0.08	0.80	0.26
Energy in milk (Mcal/d)	24.5	25.2	25.9	26.7	0.38	<0.01	0.06	0.83	<0.01	<0.01	0.03	0.16
Energy in milk/ME[4] (%)	42.5	44.6	45.4	48.8	0.91	<0.01	0.04	0.92	<0.01	<0.01	0.85	0.33
Milk urea (mg/L)	129	127	190	190	6.8	<0.01	0.90	0.85	<0.01	0.09	0.70	0.15
N efficiency (%)	30.8	32.4	33.0	32.5	0.55	0.05	0.32	0.07	<0.01	<0.01	0.46	0.40
MFP[5] (g/d)	340	344	385	400	6.7	<0.01	0.17	0.45	<0.01	<0.001	0.38	0.62
MP efficiency[6] (%)	70.6	71.8	72.2	70.5	0.73	0.89	0.71	0.06	<0.01	<0.01	0.48	0.71

[1]LSHD: low-starch, high-degradable in the rumen; HSLD: high-starch, low-degradable in the rumen; AA−: without AA supplementation; AA+: with rumen-protected AA supplies (Lys, Met, and His).

[2]Probability values correspond to the effect of the starch content (St.: low starch vs. high starch); the effect of AA supplementation (AA: AA− vs. AA+); the effect of time (T); and interactions between starch, AA, and time: St. × AA, St. × T, AA × T, St. × AA × T. When interactions between treatment × time are significant, the next line shows the corresponding slopes of the variable.

[3]In percent of true protein content.

[4]ME intake.

[5]Metabolic fecal protein estimated by INRA (2018).

[6]MP efficiency calculated in INRA (2018) using protein digestible in the intestine (PDI).

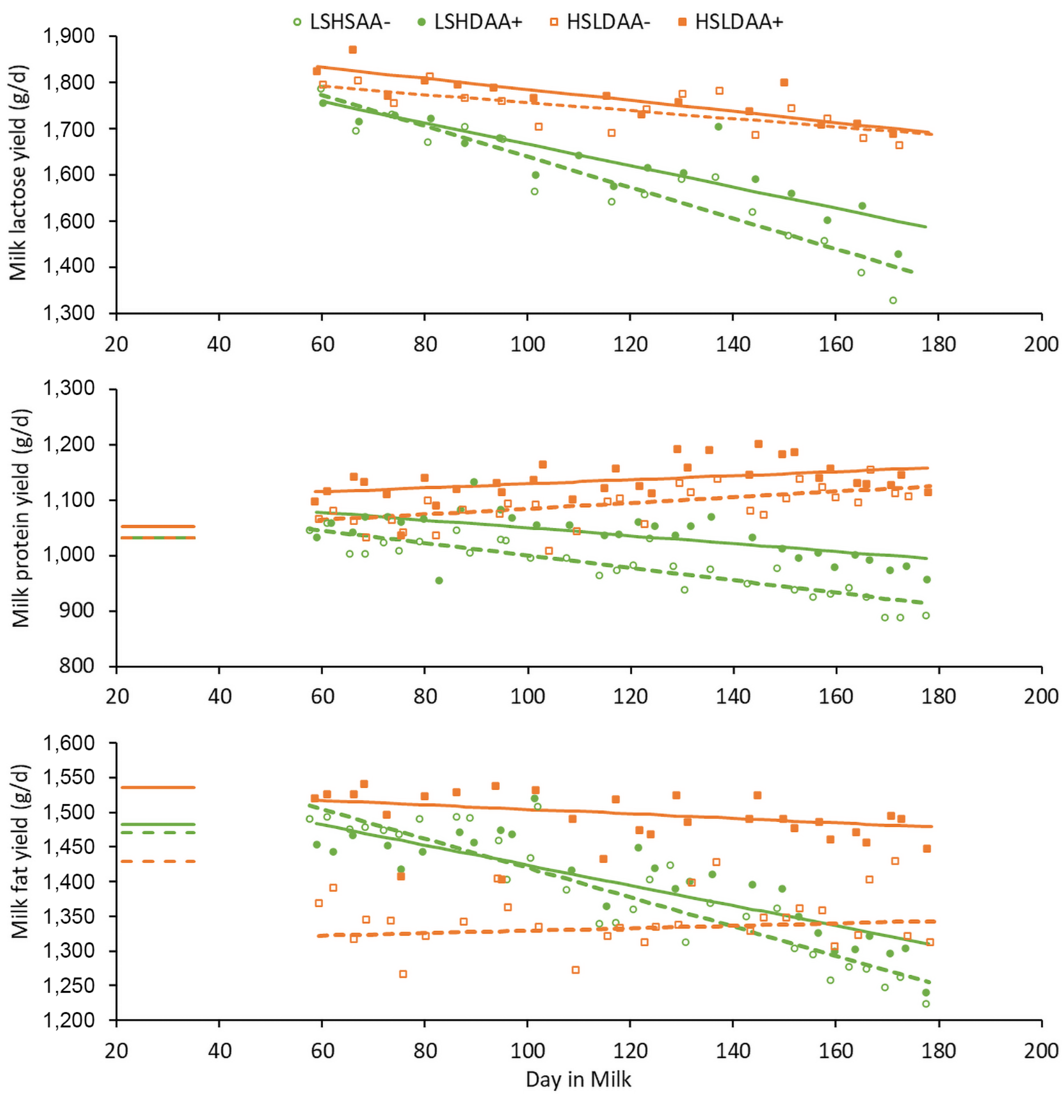

Figure 2. Effects of increasing the bypass starch content and AA supplementation on milk lactose, true protein, and fat yields from 56 to 183 DIM. The symbols represent the means of raw data per dietary treatments; open green circle = LSHDAA−, low-starch high-degradable in the rumen without AA supplementation; solid green circle = LSHDAA+, low-starch high-degradable in the rumen with rumen-protected Lys, Met, and His supplementation; open orange square = HSLDAA−, high-starch low-degradable in the rumen without AA supplementation; solid orange square = HSLDAA+, high-starch low-degradable in the rumen with rumen-protected Lys, Met, and His supplementation. Lines between 21 and 35 DIM represent covariables (dashed green line = LSHDAA−; solid green line = LSHDAA+; dashed orange line = HSLDAA−; solid orange line = HSLDAA+). From 56 to 183 DIM, the mean slopes are also represented by covariable lines (dashed green line = LSHDAA−; solid green line = LSHDAA+; dashed orange line = HSLDAA−; solid orange line = HSLDAA+). The mean of the individual slopes is not mathematically the same as the slopes presented in Tables 4 and 5, because the statistical model included the covariable correction.

in milk and ME) increased significantly with the higher starch content (43.6% vs. 47.1% on LSHD vs. HSLD diets, $P < 0.01$) and also increased with RP-AA supplementation (44.0% vs. 46.7% on AA− vs. AA+ diets; $P = 0.04$). A tendency ($P < 0.07$) to a starch × AA interaction was observed for N efficiency, which was lower in LSHDAA− than with other treatments (contrast test: LSHDAA+ vs. LSHDAA−, $P < 0.01$). With increasing starch supplies, the estimates of metabolic fecal protein (the second largest protein yield exported after milk, considered in MP efficiency by INRA, 2018) increased on the HSLD versus LSHD diets (393 vs. 342 g/d, $P < 0.01$). Milk protein yield and MP efficiency showed a starch × AA interaction, but no individual contrast test was significant. Starch × time interactions were observed on efficiencies.

Mammary Cell Proliferation and Apoptosis in Multiparous Cows

The percentage of cell proliferation in mammary tissue (measured using the double PCNA and 4′,6-diamidino-2-phenylindole staining methods) was determined in only the 32 multiparous cows (Table 6) and increased on HSLD diet versus LSHD diet and (+2.2%) with RP-AA supplementation (+4.1%), respectively ($P < 0.01$). Apoptosis rates did not vary among treatments but increased between the first and second biopsies (time effect: $P < 0.01$). The milk yield response to dietary treatments in multiparous cows was close to the milk yield response of all the cows (44, including 12 primiparous); it increased with HSLD versus LSLD diets and over time (starch × time: $P < 0.01$). The AA supplementation in multiparous cows did not increase their mean milk yield, but AA+ versus AA− diets tended to increase milk yield over time (AA × time: $P = 0.08$). Finally, lactose presented a starch × AA × time interaction in multiparous cows ($P = 0.04$), showing a lower persistency in LSHDAA− versus LSHDAA+ (contrast test: $P = 0.02$) and a higher persistency in HSLD versus LSHD diets.

Plasma Metabolites and Insulin

Energy Metabolites. Plasma glucose (Table 7) and insulin levels did not differ between the treatments. The plasma acetate concentration decreased with HSLD versus LSHD diets (−131 µmol/L, $P = 0.01$, respectively). Plasma concentration of triglycerides decreased with HSLD diets versus LSHD diets (−9.3 mg/L, $P = 0.02$). Starch × time interactions were significant, or tended to be, for plasma concentrations of BHB, acetate, triglycerides, and NEFA. These changes were due to differing time trends between the LSHD and HSLD diets during the first part of the experiment (73 DIM to 105 DIM), but subsequent time trends were similar between the LSHD and HSLD diets.

Plasma Urea and AA. Plasma urea (Table 7) increased with HSLD compared with LSHD diets (+58.5 mg/L, $P < 0.01$) and tended to decrease under RP-AA treatments (−14.5 mg/L, $P = 0.06$). Plasma Lys and Met concentrations increased with RP-AA supplementation (+9.5 and +7 µM, $P = 0.03$ and $P < 0.01$, respectively; Table 8). Plasma concentration of His exhibited a starch × AA interaction with a lower concentration of LSHDAA− than in the 3 other treatments (contrast test: $P < 0.01$). Plasma concentrations of 3-methylhistidine (**3-MH** or τ-methylhistidine for tele-methylhistidine) and Gly decreased, whereas that of Ser tended to decrease under RP-AA supplementation. Plasma concentrations of carnosine (a His-based dipeptide), Arg, Cit, and Gly levels rose with an increasing bypass starch content

Table 6. Effects of increasing the bypass starch content and AA supplementation[1] on mammary cell proliferation and apoptosis in multiparous cows and the slope of performance of multiparous cows

Item	LSHD		HSLD		SEM	P-value[2]						
	AA−	AA+	AA−	AA+		St.	AA	St. × AA	T	St. × T	AA × T	St. × AA × T
Proliferation cells[3] (%)	10.3	13.8	11.9	16.5	0.82	<0.01	<0.01	0.73	<0.01	0.24	0.32	0.81
Apoptosis cells[4] (%)	0.176	0.140	0.138	0.150	0.0208	0.49	0.55	0.20	<0.01	0.78	0.78	0.86
Milk yield (kg/d)	34.4	35.7	39.0	38.9	0.61	<0.01	0.32	0.24	<0.01	<0.01	0.08	0.30
Milk slope (g/d)	−87	−69	−37	−32	6.5							
Lactose (g/d)	1,685	1,730	1,899	1,892	37.6	<0.01	0.59	0.48	<0.01	<0.01	0.02	0.08
Lactose slope (g/d)	−4.1	−2.8	−1.5	−1.3	0.31							

[1]LSHD: low-starch, high-degradable in the rumen; HSLD: high-starch, low-degradable in the rumen; AA−: without AA supplementation; AA+: with rumen-protected AA supplies (Lys, Met, and His).

[2]Probability values correspond to the effect of the starch content (St.: low starch vs. high starch); the effect of AA supplementation (AA: AA− vs. AA+); the effect of time (T); and interactions between starch, AA and time: St. × AA, St. × T, AA × T, and St. × AA × T. When interactions between treatments × time are significant, the next line shows the corresponding slopes of the variable.

[3]Determined by immunohistochemical method staining for proliferation cell nuclear antigen (PCNA) on 2 biopsies performed at 105 and 178 DIM.

[4]Based on DNA fragmentation detection using terminal deoxynucleotidyl transferase–mediated 2′-deoxyuridine 5′-triphosphate nick-end labeling on 2 biopsies performed at 105 and 178 DIM.

Table 7. Effects of increasing the bypass starch content and AA supplementation[1] on plasma metabolites and insulin

Item	LSHD		HSLD		SEM	P-value[2]					
	AA−	AA+	AA−	AA+		St.	AA	St. × AA	T	St. × T	AA × T
Glucose (mg/L)	702	695	690	664	17.2	0.22	0.33	0.57	<0.01	0.69	1.00
BHB (μmol/L)	504	508	456	492	19.5	0.10	0.27	0.37	<0.01	<0.01	0.68
Acetate (μmol/L)	1,249	1,241	1,124	1,104	45.3	0.01	0.77	0.90	<0.01	0.03	0.30
Triglyceride (mg/L)	100.3	102.2	93.9	90	3.8	0.02	0.80	0.45	0.80	<0.01	0.03
NEFA[3] (μmol/L)	114	132	123	117	11.6	0.81	0.58	0.30	0.05	<0.01	0.86
Lactate (mmol/L)	452	433	521	403	70.3	0.80	0.28	0.43	0.13	0.35	0.19
Insulin (μU/mL)	2.78	2.81	2.98	2.90	0.361	0.67	0.95	0.87	0.93	0.93	0.08
Urea (mg/L)	127	118	191	171	7.9	<0.01	0.06	0.49	<0.01	0.01	0.42

[1]LSHD: low-starch, high-degradable in the rumen; HSLD: high-starch, low-degradable in the rumen; AA−: without AA supplementation; AA+: with rumen-protected AA supplies (Lys, Met, and His).

[2]Probability values correspond to the effect of the starch content (St.: low starch vs. high starch); the effect of AA supplementation (AA: AA− vs. AA+); the effect of time (T); and interactions between starch, AA, and time: St. × AA, St. × T, AA × T, and St. × AA × T. No starch × AA × time interaction was significant.

[3]Nonesterified fatty acids.

($P \leq 0.05$), but the plasma Ser concentration tended to decrease with the increased bypass starch content (P = 0.1). Numerous other AA also displayed a starch × AA interaction, with LSHDAA− presenting lower concentrations. This was the case for Ile, Leu, Val, and their sum (BCAA), EAA from group 2 (Lys + Ile + Leu + Val), AA from group 1 (His + Met + Phe + Tyr + Trp), and EAA. In addition, AA × time interactions tended to be observed for His, Ile, Leu, Val, BCAA, and EAA from group 2, which corresponded to a progressive decrease over time in their concentrations under the LSHDAA− diet, whereas they remained more stable with the 3 other treatments. Aspartic acid and Cys also showed starch × AA interactions with lower concentrations on the LSHDAA− diet than the LSHDAA+ diet (contrast test: $P = 0.03$ for both). The starch × AA interactions for Phe, Trp, Ala, Asn, Gln, Orn, Tyr, 1-methylhistidine (1-MH or π-methylhistidine), NEAA, and total AA (**TAA**) were explained by the lower plasma concentrations (significant or trend) in the HSLDAA+ diet than in the HSLDAA− diet (all contrast tests: $P \leq 0.07$).

DISCUSSION

This study investigated the effects of increasing bypass starch intake and supplementing with RP-Lys, RP-Met, and RP-His on lactation persistency, milk yield and composition, plasma metabolites, mammary cell proliferation, and apoptosis from 56 to 183 DIM. We studied lactation persistency through the estimated slope of milk decline. Although the literature reports that AA exert a stronger positive effect on milk yield during the transition period than in mid lactation, in this experiment we chose to focus on the effect of these dietary treatments on lactation persistency, that is, after the peak of lactation. This approach also enabled us to use a reference period to balance the experimental groups and estimate the covariates. With the limited number of cows (4 × 11 per treatment), the covariate appears essential to correct allocation biases. Consistent with field observations in European dairy systems and with reports of increased energy partitioning toward body fat in response to higher bypass starch diets (Boerman et al., 2015; NASEM, 2021), we hypothesized that a greater bypass starch supply may modify energy partitioning through an increased BCS. We further hypothesized that this higher partitioning toward body reserves would be reflected in a decrease in lactation persistence for higher bypass starch diets. We also hypothesized a more pronounced increase in milk and protein yields in response to supplies of Lys, Met, and His at INRA (2018) requirements at the beginning of the experiment than at the end. All the diets were relatively low in MP supply, as planned (<95 g/kg DM of PDI). Although we formulated the diets to supply the same MP and NEL contents, the HSLD diet had a slightly higher MP content (+3.8 g/kg DM of PDI through RUP) based on INRA (2018). This difference in MP content was mainly attributed to the higher RUP contents of the HSLD concentrate and corn silage than initially estimated before the experiment, whereas the RUP content of grass haylage in LSHD was lower than expected. In addition, NEL content (−0.085 Mcal/kg DM; calculated using OMD in INRA, 2018) was also lower in HSLD versus LSHD diets because grass haylage in LSHD provided more NEL and corn silages provided less NEL than expected. These slight changes in MP and NEL contents could influence the production response of the animals. First, we will analyze the production response in relation to possible changes in nutrient flows due to digestion. Then we will analyze indicators of nutrient partitioning and mammary gland physiology.

Table 8. Effects of increasing the bypass starch content and AA supplementation[1] on plasma AA and AA derivatives

	LSHD		HSLD			*P*-value[2]					
AA, µ*M*	AA−	AA+	AA−	AA+	SEM	St.	AA	St. × AA	T	St. × T	AA × T
Total AA	1,996	2,113	2,133	1908	57.9	0.57	0.36	0.01	0.34	0.31	0.09
EAA	657	768	762	730	26.4	0.22	0.15	0.01	0.45	0.27	0.11
Arg	50.3	60.5	62.1	59.3	2.58	0.05	0.16	0.23	0.26	0.63	0.23
His	24.9	36.3	43.5	38.9	3.41	<0.01	0.33	0.03	0.31	0.61	0.16
Ile	76.9	92.0	92.6	86.6	3.59	0.16	0.22	0.01	0.10	0.11	0.07
Leu	106	121	123	113	4.91	0.37	0.61	0.02	0.46	0.09	0.06
Lys	58.1	74.1	66.9	69.9	4.02	0.57	0.03	0.12	0.37	0.48	0.31
Met	20.4	29.5	22.4	27.3	1.47	0.94	<0.01	0.16	0.76	0.19	0.62
Phe	41.1	41.8	41.1	36.5	1.49	0.09	0.19	0.08	0.61	0.25	0.17
Thr	92.4	103.2	102.7	102.1	4.96	0.36	0.30	0.26	<0.01	0.26	0.12
Trp	35.0	37.8	36.6	32.5	1.30	0.16	0.62	0.01	<0.01	0.20	0.16
Val	152	172	171	163	6.49	0.43	0.32	0.04	0.51	0.13	0.06
NEAA	1,338	1,345	1,371	1,178	40.3	0.11	0.03	0.02	0.19	0.30	0.11
Ala	244	258	279	257	10.5	0.12	0.69	0.09	<0.01	0.15	0.10
Asn	46.1	49.9	55.9	49.1	1.81	0.02	0.41	<0.01	0.45	0.64	0.21
Asp	3.85	4.51	4.51	4.06	0.278	0.71	0.71	0.05	0.21	0.62	0.09
Cys	7.29	9.3	10.11	9.75	0.577	0.01	0.17	0.05	<0.01	0.84	0.20
Gln	244	257	277	236	12.3	0.64	0.25	0.04	0.12	0.43	0.33
Glu	37.0	39.3	39.4	35.1	2.04	0.68	0.63	0.11	0.28	0.28	0.02
Gly	490	455	432	352	20.3	<0.01	0.01	0.28	0.08	0.14	0.20
Pro	97.1	103.3	103.1	99.1	4.47	0.84	0.80	0.26	0.73	0.58	0.07
Ser	123.4	121.2	121.3	94.8	8.36	0.10	0.10	0.16	0.13	0.43	0.29
Tyr	44.8	47.5	48.4	42.4	2.28	0.74	0.47	0.06	0.64	0.16	0.50
Nonproteinogenic AA											
Cit	66.9	66.9	78.6	77.3	4.69	0.02	0.89	0.88	0.01	0.64	0.02
Orn	33.0	37.4	37.3	32.4	2.35	0.88	0.92	0.06	0.17	0.27	0.11
Tau	43.4	47.6	45.8	48.6	3.28	0.60	0.29	0.82	0.02	0.16	0.23
Sums											
BCAA[3]	335	385	387	363	14.6	0.32	0.36	0.02	0.36	0.11	0.06
AA group 1[4]	166	192	192	178	7.6	0.46	0.46	0.01	0.79	0.45	0.31
EAA group 2[5]	393	460	454	433	18.0	0.35	0.21	0.02	0.38	0.16	0.08
Derivatives											
Carnosine	14.4	14.6	20.1	18.7	1.31	<0.01	0.64	0.54	<0.01	0.90	0.18
3-MH[6]	5.65	5.30	7.17	5.72	0.335	0.01	0.01	0.11	0.04	0.51	0.31
1-MH[7]	4.57	4.72	5.76	4.53	0.322	0.13	0.10	0.04	0.27	0.15	0.10

[1]LSHD: low-starch, high-degradable in the rumen; HSLD: high-starch, low-degradable in the rumen; AA−: without AA supplementation; AA+: with rumen-protected AA supplies (Lys, Met, and His).

[2]Probability values correspond to the effect of the starch content (St.: low starch vs. high starch); the effect of AA supplementation (AA: AA− vs. AA+) and the time (T) effect; and the interactions between starch and AA (St. × AA). No starch × AA × time interaction was significant.

[3]Ile + Leu + Val.

[4]His + Met + Phe + Tyr + Trp.

[5]Ile + Lys + Leu + Val.

[6]3-Methylhistidine; i.e., τ-methylhistidine; CAS no. 332 80-9.

[7]1-Methylhistdine; i.e., π-methylhistidine; CAS no. 368 16-1.

High Bypass Starch Diet

Increasing Bypass Starch Increased Milk and Protein Yields. The increased MP content, together with the higher starch content, may have contributed to the greater MPY observed in the HSLD diet compared with the LSHD diet. The consistently higher plasma urea concentration could suggest an increased MP supply and AA catabolism in the HSLD diet compared with the LSDH diet. However, this result was not consistent with the fact that HSLD treatments did not cause an increase in plasma TAA, EAA, and NEAA levels or a loss of N efficiency. By contrast, N efficiencies increased under the 2 HSLD treatments. Plasma urea concentration could also have increased because urea added to the HSLD diet to equilibrate the rumen protein balance was not well used by the rumen microbes. In this hypothesis, the microbial protein would have been lower in HSLD diet than predicted, leading to no increase in MP supply. In addition, the lower NEL content in HSLD diet may therefore have counteracted the potential effect of additional MP content on MPY in accordance with Daniel et al. (2016). Given the observed DMI, INRA (2018) predicted no change in MPY or MY, as the positive marginal response of MPY to the increase in MP intake (+154 g/d) was offset by the negative marginal response to the concomitant decrease in NEL intake (−1.5 Mcal/d) within the model. Increases in milk or protein yields, or both, had been observed in several studies that increased the starch content within the range of 184 to 322 g/kg DM, mainly by including corn grain in the concentrate (Oba and Allen, 2003; Akins et al., 2014; Silvestre et al., 2022), as well as in reviews or meta-analyses (Nocek and Tamminga, 1991; Ferraretto et al., 2013). Rumen propionate supply or duodenum glucose infusions increase whole body glucose rate of appearance (i.e., sum of glucose synthesis, glucose absorption, and glycogenolysis; Rigout et al., 2002; Lemosquet et al., 2009). Rumen propionate infusion and postrumen starch or glucose infusion also increased milk and milk protein yields (Reynolds et al., 2001; Rigout et al., 2003). In our experiment, the INRA (2018) system predicted lower ruminal VFA production but greater small-intestinal glucose absorption with HSLD compared with LSHD diets (Table 4), based on meta-analyses by Offner and Sauvant (2004), Nozière et al. (2011), and Loncke et al. (2009). The system also predicted greater hindgut starch digestion (238 vs. 72 g/d) and starch in feces (258 vs. 106 g/d) with HSLD versus LSHD diets, consistent with Reynolds et al. (2001). Hindgut VFA were included in ED and NEL estimations through the DOM (INRA, 2018). However, uncertainties remain regarding small intestine starch and hindgut starch digestions (Moharrery et al., 2014). The absence of an increase in plasma glucose concentration with HSLD versus LSHD could suggest limitations in small-intestinal glucose absorption or greater glucose utilization by portal-drained viscera. Nevertheless, Loncke et al. (2009) predicted a higher net portal appearance of glucose with HSLD (−647 vs. 563 g/d) than with LSHD diet. Consistent with this prediction, we observed a significant increase in lactose yield (+153 g/d) with HSLD, suggesting a greater mammary glucose uptake in conjunction with a higher whole body glucose appearance rate.

Higher Lactation Persistency with Bypass Starch. The higher bypass starch content increased the lactation persistency (Figure 1) and the persistency of milk component yields (starch × time interactions). These effects on lactation persistency were observed irrespective of the level of AA in the diet and were associated with a trend toward higher DMI slope. Interestingly, mean DMI did not differ significantly with HSLD than with LSHD, although mean milk yield increased on HSLD versus LSHD diets. This might suggest that the higher milk yield in HSLD versus LSHD diets was driving this DMI increase. However, the gradual increase in DMI probably sustained this greater persistence of lactation. To our knowledge, lactation persistency has only been reported in 2 studies examining different starch diets (Chen et al., 2016; van Hoeij et al., 2017), neither of which observed changes in lactation persistency in response to a higher starch content. However, it is worth noting that Chen et al. (2016) supplied a glucogenic diet at a different stage of lactation (−10 to 100 DIM) compared with our experiment (56 to 183 DIM). In addition, our study used a higher starch content (234 g/kg DM vs. 199 g/kg DM) than van Hoeij (2017). Several studies have examined the interaction between starch and time (Akins et al., 2014; Piccioli-Cappelli et al., 2014). Akins et al. (2014) started to apply their treatments later in lactation and for a shorter period than us (90–174 DIM); they did not observe any interactions between starch and time for milk yield and MPY but did find one for fat yield. Similarly, we found a starch × time interaction, with the slope of MFY declining less steeply on HSLD diets than on LSHD diets. In this context, Piccioli-Cappelli et al. (2014) showed an increased effect of starch on milk, lactose, and protein yields in late lactation but not in early lactation. This is in line with our results because increasing the starch content clearly but gradually modified milk synthesis.

An Increase in Metabolic Energy Partitioning into Milk with Bypass Starch. Interestingly, energy efficiency (GE exported in milk/ME intake) increased with a higher bypass starch content, as seen by Zang et al. (2021), although they observed a decrease in DMI under high-starch treatment. This higher energy efficiency in HSLD versus LSHD diets was in accordance with the site of starch digestion, as ruminally fermented starch is less energy-efficient than starch digested in the small intestine (Harmon and McLeod, 2001). Contrary to several experiments, which increased the dietary starch content (Reynolds et al., 2001; Piccioli-Cappelli et al., 2014; Boerman et al., 2015), and to our initial hypothesis, we did not find any indication of important changes to energy partitioning that favored over-fattening at the end of lactation, as in van Hoej et al. (2017). This could partially be explained by the lower starch content of our HSLD diet compared to that used by Boerman et al. (2015) and Oba and Allen (2003). The other reason might be that our experiment ended when cows were only at 183 DIM, insofar as we observed a tendency toward a small BW change (tendency toward higher slopes in HSLD vs. LSHD treatments, Table 5) that might increase in later lactation. Another probable difference between the present experiment and others regarding energy partitioning was because the ME and NEL content and intake were slightly lower in HSLD than in LSHD diets. Indeed, marginal NEL efficiency for milk increased when NEL supplies were reduced (Daniel et al., 2016), leading to a higher proportion of NEL exported in milk. We did not observe any differences in BCS, plasma glucose, and insulin concentrations in cows receiving HSLD versus LSHD treatments throughout the experiment. In addition, a greater persistency of milk and milk components was associated with higher mammary cell proliferation in multiparous cows receiving the 2 HSLD treatments. All these measurements therefore strongly suggest that in response to the increase in starch content without increasing NEL content (HSLD vs. LSHD), energy was driven more toward the mammary gland to synthesize more milk.

Rumen-Protected AA Supplementation: Effects and Interactions

Effect of RP-AA on Milk Protein Yield. Supplementing with Lys, Met, and His increased MPY and tended to increase milk true protein content, as expected. However, the effect observed (+37 g/d) was less than the predicted increase in MPY (+50 g/d) in AA+ versus AA− diets according to INRA (2018), taking account of differences in MP, NEL, LysDI, and MetDI. We also expected a stronger MPY response to AA+ versus AA− because we also increased the supply of His. Nevertheless, in 3 experiments using the same 3 RP-AA (Zang et al., 2021; Van den Bossche et al., 2023; Nichols et al., 2024) as us, 2 found a similar increase in milk protein content (Zang et al., 2021; Nichols et al., 2024).

Contrary to our second hypothesis, supplementation with Lys, Met, and His increased MPY and content without any RP-AA × time interaction. Our results differed from those of Socha et al. (2008), who showed a weaker MPY response to a postrumen infusion of Met administered to the cows at the end of lactation rather than at the beginning of lactation. However, Carder and Weiss (2017) reported carry-over effects (from 23 to 63 DIM) with increases in protein and fat content after stopping RP-Met supplementation (performed between 3 and 23 DIM). It cannot be excluded that RP-AA supplementation in the first part of the present study (e.g., 56 to 120 DIM) induced a carry-over effect on the second part (from 121 to 183 DIM), additive to a weaker MPY effect due to the stage of lactation. This may have contributed to maintaining a constant MPY response throughout the experiment.

Rumen-Protected Amino Acids Increased MFY. With RP-AA supplementation, we observed a tendency toward an increase in MFY and an increase in the quantity of energy exported in milk, as observed by Giallongo et al. (2016). Increases in ECM and ECM/DMI were reported with a higher Lys+Met through RP-AA or with RP-Lys alone (Socha et al., 2005; meta-analysis by Arshad et al., 2024). The increase in MFY may have resulted from an imbalance in Lys, Met, His, or EAA supplies (Cant et al., 2003; Nichols et al., 2024). However, we supplied less RP-AA than Nichols et al. (2024), thus creating a less marked imbalance. These 3 AA may also play key roles in the ability of the mammary gland to export energy in milk through the stimulation of pathways involved in milk component synthesis (Li et al., 2016; Nichols et al., 2020).

No Interaction of RP-AA with Starch Supplies Affecting Milk Protein Yield. We observed an additive effect on MPY when increasing the bypass starch content of diets and supplementing with Lys, Met, and His. The additive effects on MPY of MP × NEL (Daniel et al., 2016) or digestible energy and AA supplies (Hanigan et al., 2024) had previously been reported in meta-analyses. Additive effects were also reported in mammary metabolism studies on increasing starch or intestinal glucose × MP (Cantalapiedra-Hijar et al., 2015; Nichols et al., 2019; Omphalius et al., 2020), except in Rius et al. (2010). In addition, increasing the 3-RP-AA supply lowered plasma urea concentrations, probably through a decrease in AA catabolism linked to a higher utilization of AA in milk protein synthesis as expected (Haque et al., 2012). However, in our study, this RP-AA supply only increased N efficiency with the LSHD diet but not the HSLD diet, while MP efficiency did not change. These absences of increased efficiencies were explained by the smaller increase in MPY than those seen in CP or MP intakes with AA+ versus AA− treatments, as in Socha et al. (2008).

Interactions Between RP-AA and Starch Supplies and Lactation Persistency. Interestingly, many plasma AA concentrations exhibited an interaction between starch and AA. This suggests that RP-AA supplementation modified the metabolism of these AA differently depending on the starch level (or MP intake, or both) to increase MPY in an additive manner. Lower plasma NEAA concentrations were observed in the HSLDAA+ treatment compared with the others, suggesting a change in their metabolism. Conversely, the LSHDAA− treatment had lower plasma concentrations of His, Ile, Leu, and Val than the other treatments. The lower plasma His concentration suggested that His intake was limiting only in the LSHDAA− treatment. Indeed, the estimated HisDI intake (g/d; Table 4; INRA, 2018) increased (+10%) slightly more than LysDI (+8%) and MetDI (+3%) intakes with HSLDAA− versus LSHDAA−, although no significant interaction was observed. The HSLDAA− diet supplied more HisDI than the LSHDAA− diet, due to a higher supply of MP combined with a lower proportion of MP derived from rumen microbial protein, as microbial protein is a relatively poor source of His (Schwab and Broderick, 2017). In addition, His metabolism appeared to be altered by the HSLD treatments because carnosine and His peptide, as well as the methylated His derivatives 3-MH and 1-MH, increased with HSLD versus LSHD diet, with an interaction observed only for 1-MH. These changes could indicate a greater protein mobilization but likely a greater whole body protein turnover under this high-starch, high-MP condition (i.e., HSLD diet). A higher protein turnover was previously observed with increasing rumen propionate infusion (Raggio et al., 2006). This increased turnover could also contribute to the higher plasma concentration of urea in HSLD cows compared with LSHD cows, as some AA and AA derivatives, such as methyl derivatives of His, could not be recycled in the turnover and could be catabolized. Nichols et al. (2016) also demonstrated that postrumen glucose infusion increased plasma urea concentration. However, the increase in Nichols et al. (2016) corresponded to a higher increase in EAA supply than in the 2 HSLD treatments.

Supplementing with RP-AA decreased the concentration of 3-MH, a body protein turnover marker derived from His, suggesting lower AA catabolism and preferential use of absorbed AA for productive functions, notably milk protein synthesis. The reduction (trend) in plasma urea with AA+ versus AA− across both starch levels further supports a decrease in AA catabolism. The plasma concentration of BCAA decreased over time in LSHDAA− versus other treatments. These changes in plasma BCAA could not be explained by the increase in starch because post rumen glucose supply is well known to decrease BCAA (Omphalius et al., 2020). The increase in plasma BCAA concentration could then reflect the correction of limiting EAA (Lys, Met, and His) in LSHDAA+ and in the 2 HSLD treatments because MP intake increased. Our results are consistent with those of Giallongo et al. (2017), who observed decreases in plasma BCAA over time when comparing a His-deficient diet to one with higher MP and adequate His supplies. Interestingly, they reported increased milk yield and DMI under higher MP and an adequate His supply as our 2 HSLD treatments.

We also observed a trend toward an interaction affecting both lactation persistency and DMI (starch × AA × time). Supplementation with RP-AA did not increase mean DMI and milk yield in accordance with studies using the same RP-AA (Zang et al., 2021; Van den Bossche et al., 2023; Nichols et al., 2024). However, RP-AA supplementation of the LSHD diet was associated with an increased lactation persistency (higher slopes) of lactose, fat, and total milk secretions and DMI (slices for LSHDAA− vs. LSHDAA+; starch × AA × time interaction or tendencies toward interactions). All this suggests that lactation persistency is sensitive to AA supply, and perhaps especially to His, because His is the AA that has been reported in the literature (Vanhatalo et al., 1999; Lee et al., 2012; Räisänen et al., 2023) to increase milk yield and DMI in mid-lactating cows. The absence of the effects of RP-AA, particularly His, on lactation persistency in the HSLD diet may be due to lower microbial protein synthesis than predicted by INRA (2018). The higher plasma and milk urea concentrations in HSLD versus LSHD diets might suggest that dietary urea was not effectively utilized in rumen, resulting in lower microbial protein synthesis. In fact, His is deficient in diets where microbial protein represents a high proportion of the MP supply (Schwab and Broderick, 2017). In addition, the supply and metabolism of His were also altered in the HSLD treatments, which could induce a higher lactation persistency than the LSHD treatments. We can hypothesize that the higher lactation persistency in the present experiment resulted both from a higher His supply in LSHDAA+ versus LSHDAA− and from higher His and starch supplies in HSLD versus LSHD diets.

Mammary Cell Proliferation and Lactation Persistency

Several mechanisms at the mammary gland level could explain the changes in milk secretions observed in the present experiment. First, in terms of metabolism, considering that HSLD effectively increased whole body glucose availability, it has already been reported that glucose and EAA increased milk protein yield additively (Omphalius et al., 2020). Glucose decreased the mammary catabolism of AA from group 2 catabolism (Ile, Leu, Lys, and Val), which were reduced in AA plus glucose supply compared with AA supply alone. This enabled a greater mammary exportation of those AA in milk protein (Meta-analysis of Lemosquet et al., 2010; Omphalius et al., 2020). In addition, glucose and AA modified signaling pathways, particularly the mammalian target of rapamycin (mTOR), which regulates milk protein (Toerien et al., 2010; Arriola Apelo et al., 2014; Wang et al., 2019) and fat synthesis (Li et al., 2016; Nichols et al., 2020). In the present experiment, we chose to analyze mammary cell apoptosis and proliferation in multiparous cows for several reasons. First, changes in the number of secretory cells could be related to lactation persistency (Capuco et al., 2003). Second, the stimulation of cell proliferation in the mammary tissue was observed in cows fed diets with higher energy densities (Nørgaard et al., 2005). Third, higher expression of the MYC gene, which is involved in cell proliferation, was observed with an increased postrumen glucose supply (Nichols et al., 2020). Finally, the effects of AA supplementation on indicators of cell turnover in the mammary gland have already been observed in dairy ruminants (Nichols et al., 2017; Boutinaud et al., 2020). We observed an increased cell proliferation in the mammary tissue of multiparous cows in response to both RP-AA and increased bypass starch supplies, as indicated by PCNA markers, with no effect on mammary cell apoptosis. However, these increases in cell proliferation occurred without any starch × AA interaction, while lactation persistency tended to be higher in LSHDAA+ versus LSHDAA− and was still higher in both HSLD treatments compared with both LSHD treatments. One difference between increasing the bypass starch supply and RP-AA supplementation was the increase in DMI with time with bypass starch, which probably supported lactation persistency through a greater supply of all nutrients to the mammary gland.

CONCLUSIONS

The high bypass starch diet yielded the highest milk and protein yields and, contrary to our first hypothesis, led to improved lactation persistency. The slightly lower supplies of ME and NEL in the HSLD versus LSHD diets may have favored this increased lactation persistency and the export of milk components rather than increasing BCS. The main effect of AA supplementation was increases in milk protein and fat yields and milk protein content (trend). In contrast to our second hypothesis, these responses did not decline as lactation progressed. In addition, the lowest lactation persistency was observed with the diet with the low-starch content not supplemented with RP-Lys, Met, and His (LSHDAA−), highlighting the importance of nutrient supplies (starch or AA) in diets with relatively low-MP contents (<95 g/kg DM). Further research is required to better understand the regulation of lactation persistency and the involvement of starch and AA, especially of His, and mammary cell proliferation.

NOTES

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Nonstandard abbreviations used: 1-MH = 1-methylhistidine (i.e., π-methylhistidine); 3-MH = 3-methylhistidine (i.e., τ-methylhistidine); AA− = without Lys, Met, and His supply; AA+ = with RP-Lys, RP-Met, and RP-His supply; AADI = AA digestible in the intestine; AlaDI = Ala digestible in the intestine; ArgDI = Arg digestible in the intestine; AspDI = Asp digestible in the intestine; BCAA = branched-chain AA; DNDF = digestible NDF; DOM = digestible OM; ED6_N = effective degradability coefficients for N, calculated for a passage rate of 6% per hour; ED6_st = effective degradability coefficients for starch, calculated for a passage rate of 6% per hour; FOM = fermentable OM; GE = gross energy; GluDI = Glu digestible in the intestine; GlyDI = Gly digestible in the intestine; HisDI = His digestible in the intestine; HSLD = high-starch, low-degradable in the rumen; HSLDAA− = HSLD without AA supplementation; HSLDAA+ = HSLD with AA supplementation; IleDI = Ile digestible in the intestine; LeuDI = Leu digestible in the intestine; LSHD = low-starch, high-degradable in the rumen; LSHDAA− = LSHD without AA supplementation; LSHDAA+ = LSHD with AA supplementation; LysDI = Lys digestible in the intestine; MetDI = Met digestible in the intestine; MFP = metabolic fecal protein; MFY = milk fat yield; MPY = milk true protein yield; MW = molecular weight; NCN = noncasein nitrogen; NEFA = nonesterified fatty acids; PCNA = proliferation cell nuclear antigen; PCO = proportion of concentrate; PDI = protein digestible in the intestine; PDIA = true protein digested in the small intestine from alimentary (feedstuffs; i.e., digestible RUP); PDIM = protein digestible in the intestine from microbes; PheDI = Phe digestible in the intestine; ProDI = Pro digestible in the intestine; RP-AA = rumen-protected AA; SerDI = Ser digestible in the intestine; St. = starch content; Std_si = starch digestibility in small intestine; TAA = total AA; ThrDI = Thr digestible in the intestine; TVFA = total VFA; TyrDI = Tyr digestible in the intestine; ValDI = Val digestible in the intestine.

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